

Stochastic Optimization - Extra Credit Problems

DISCLAIMER: The exercises in this document are **NOT** mandatory, and are for extra credit (up to 15 additional points). These exercises were removed from the original iteration of the project outline as they are focused on specific mathematical details, which can help provide a deeper understanding to the underlying theory behind the success of the algorithm, but are not necessary to gain a general intuition.

Background

The classical root finding problem is finding the solution for the following equation given arbitrary function f :

$$f(\theta) = 0 \quad (0.1)$$

Finding the minimum here can be done with a variety of methods: Bisection method, False position method, Secant method, Newton-Raphson method, etc.

Now suppose that we do not have f , but rather we have a noisy version of f known as g . More precisely, we define g as follows:

$$g(\theta) = f(\theta) + z, \quad E[z] = 0 \quad (0.2)$$

To reiterate, g acts as an estimate for f , but due to random noise, the values we see are slightly off. The method by which we aim to find the root of f given only g is called Stochastic Approximation.

0.1 Stochastic Approximation in simple setting

We will make the following simplifying assumptions. The algorithm needs a sequence of **positive learning rates** that we denote as $\{\eta_t\}_{t \geq 1}$.

1. The function f has a unique root θ^* (i.e., $f(\theta^*) = 0$ for a unique θ^*). This unique root is a positive zero crossing of f . In other words:

$$f(\theta^*) = 0, \quad \theta > \theta^* \Rightarrow f(\theta) > 0, \quad \theta < \theta^* \Rightarrow f(\theta) < 0$$

2. g has a finite upper and lower bound. In other words, we have bounded noise:

$$P(|g| < c) = 1 \text{ where } c \geq 0 \quad (0.3)$$

3. The noise is independent of θ :

$$\forall \theta : E[g] = f(\theta), \quad P(z|\theta) = P(z) \quad (0.4)$$

4. The **learning rates** η_t do not approach 0 too quickly or too slowly. More formally:

$$\sum_{t=1}^{\infty} \eta_t = \infty, \quad \sum_{t=1}^{\infty} \eta_t^2 = c \quad (0.5)$$

For some positive c . In other words, the sum of the learning rates is unbounded, but the sum of their squares is bounded. An example sequence that satisfies these constraints is $\eta_t = t^{-0.5}, t \geq 1$

The algorithm is defined as follows, where Θ_t is the t^{th} approximation of θ^* :

- Let Θ_1 be some initial value or guess
- For $t = 1, 2, \dots$ perform the following iteration

$$\Theta_{t+1} = \Theta_t - \eta_t G_t \quad (0.6)$$

Where $G_t = f(\Theta_t) + Z_t$, just as mentioned previously.

1 Stochastic Approximation

1.1 Convergence proof of SA under simplified settings

Define the Mean Squared Error at step t as follows

$$e_t^2 = E\{(\Theta_t - \theta^*)^2\} \quad (1.1)$$

We will prove this value converges to 0 as the SA algorithm executes.

Extra Credit Ex. 1.

1. **(2 points)** Prove the following relationship for any two random variables u, v :

$$E_u[f(u)] = E_v[E_{u|v}[f(u)|v]] \quad (1.2)$$

Do not assume any kind of independence. It may help to consider the following simplified relation first: $E[A] = E[E[A|B]]$.

Hint: You will need to have a minimal understanding of conditional expectation ($E_{u|v}[f(u)|v]$). Here ([blogpost](#) or [video](#)) are some resources to learn about conditional expectation.

2. **(0.5 point)** Using the last part, and the unbiasedness and independence assumption presented in Eq 0.4, Prove the following recurrence relation:

$$e_{t+1}^2 = e_t^2 - 2\eta_t \rho_t + \eta_t^2 E\{G_t^2\} \quad (1.3)$$

where $\rho_t := E\{(\Theta_t - \theta^*)f(\Theta_t)\}$.

3. **(0.5 point)** Unfold the recurrence relation of last part to get the following:

$$e_{t+1}^2 = e_1^2 - 2 \sum_{i=1}^t \eta_i \rho_i + \sum_{i=1}^t \eta_i^2 E[G_i^2] \quad (1.4)$$

hint: Unfolding means turning a recurrence relation of type $f(\theta_t, \theta_{t-1}) = 0$ to a relation of type $f(\theta_t, \theta_0) = 0$. Usually if you have a $f(\theta_t, \theta_{t-1}) = \theta_t - \theta_{t-1} - c_t = 0$, you can obtain $\theta_t = \theta_0 + \sum_i c_i$ by summing over the telescopic differences $\theta_i - \theta_{i-1}$.

4. **(0.5 point)** Using the bounded noise assumption, prove that $E[G_t^2]$ is bounded.
5. **(0.5 point)** Prove that $\sum_{i=1}^t \eta_i^2 E[G_i^2]$ is bounded.
6. **(0.5 point)** Use the first assumption 1 to prove that ρ_t is non-negative.
7. **(0.5 point)** Using the last two parts, and the unfolded relation 1.4, Prove the following

$$0 \leq \sum_{i=1}^t \eta_i \rho_i \leq C < \infty \quad (1.5)$$

and write down a valid bound C that does not depend on t .

8. **(0.5 point)** Letting $t \rightarrow \infty$, the last part and 1.4 jointly guarantee that the sequence e_t converges to a finite number. Does this guarantee that the sequence e_t converges to zero?
9. **(1.5 points)** Define $b_t := |\Theta_1| + c \sum_{i=1}^t \eta_i$, and $d_t := \min_{|\theta| < b_t} \frac{f(\theta)}{\theta - \theta^*}$. Prove the following three properties:

$$P\{|\Theta_t| < b_t\} = 1 \quad (1.6)$$

$$t \geq 1 : d_t > 0 \quad (1.7)$$

$$t \geq 1 : \rho_t \geq d_t e_t^2 \quad (1.8)$$

Also, it is straightforward to prove the following property, but you do not need to prove it. You can refer to and use this property in later questions.

$$\sum_{i=1}^{\infty} \eta_i d_i = \infty \quad (1.9)$$

10. **(0.5 point)** Using the properties in the last part prove that $\sum_{i=1}^{\infty} \eta_i d_i e_i^2 < \infty$.
11. **(1.5 point)** Using the evidence you have collected, finalize the proof with the last piece:

$$\lim_{n \rightarrow \infty} e_t^2 = 0$$

1.2 Simple example of SA

Let us assume the following simplified setting

- Assume $f(\theta) = \theta$. This means that the root is $\theta^* = 0$.
- Assume that $\eta_i < 1$ for all i .
- Assume that Z_i has a mean of zero and variance of 1.

$$\forall i : E[Z_i] = 0, E[Z_i^2] = 1$$

Extra Credit Ex. 2.

1. **(0.5 point)** Prove that

$$e_{t+1}^2 = (1 - \eta_t)^2 e_t^2 + \eta_t^2 \quad (1.10)$$

2. **(0.5 point)** Define u_t as follows:

$$t \geq 2 : u_t = \frac{e_t^2}{\prod_{i=1}^{t-1} (1 - \eta_i)^2} \quad (1.11)$$

and $u_1 = e_1^2$. Prove the following:

$$u_{t+1} = u_t + \frac{\eta_t^2}{\prod_{i=1}^t (1 - \eta_i)^2} \quad (1.12)$$

3. **(0.5 point)** Unfold the previous recurrence relation and prove that

$$u_{t+1} = u_1 + \sum_{j=1}^t \frac{\eta_j^2}{\prod_{i=1}^j (1 - \eta_i)^2} \quad (1.13)$$

4. **(0.5 point)** Prove the following closed form solution for e_t^2 :

$$e_t^2 = \left(\prod_{i=1}^{t-1} (1 - \eta_i)^2 \right) \left(e_1^2 + \sum_{j=1}^{t-1} \frac{\eta_j^2}{\prod_{i=1}^j (1 - \eta_i)^2} \right) \quad (1.14)$$

Stochastic Optimization Implementation - Extra Credit

Extra Credit Ex. 3.

(4 points) [This](#) is an article about different variants of SGD. Please read the article and think about the possible advantages or negative consequences of the algorithmic decisions made in each variant. These can essentially be the same weaknesses or strengths that cause a variant to over-perform or under-perform under practical data. You do not need to submit anything as a proof of reading.

Implement ADAGRAD and another variant that is listed in the above article in Exercise 4. When you implement the two variants, try to artificially make them perform better/worse than others by tuning parameters. Include some discussion or explanation as well as your plots.